Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let f be a continuous function satisfying $f(x+y)=f(x)f(y)(x,y\in\mathbf{R}$ with $f(1)=e$ then the value of $\int\frac{xf(x)}{\sqrt{1+f(x)}}dx$ is

(A) $2x\sqrt{1+e^x}-4\sqrt{1+e^x}-2\log\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right|+C$

(B) $2\sqrt{1+e^x}-4\sqrt{1+f(x)}$
$-2\log|\sqrt{1+f(x)}-1|+C$

(C) $2\sqrt{1+f(x)}-4\sqrt{1+f(x)}-2\log\left|\frac{\sqrt{1+f(x)}-1}{\sqrt{1+f(x)}+1}\right|+C$

(D) none of these

Q.2

Let f be a continuous function satisfying $f(x+y)=f(x)+f(y)$, for each $x,y\in\mathbf{R}$ and $f(1)=2$ then $\int\frac{f(x)\tan^{-1}x}{(1+x^2)^2}\,\mathrm{d}x$ is equal to

(A) cannot be determined explicitly

(B) $C-\frac{\tan^{-1}x}{2(1+x^2)}+\frac{1}{4}\tan^{-1}x+\frac{f(x)}{1+(f(x))^2}$

(C) $C-\frac{\tan x}{2(1+x^2)}+\frac{\tan^{-1}x}{4}+\frac{x}{4(1+x^2)}$

(D) $C-\frac{1}{(1+x^2)}\tan^{-1}x+\frac{1}{2}\tan^{-1}x+\frac{x}{2(1+x^2)}$

Q.3

If $\int\frac{f(x)}{x^2-x+1}\,\mathrm{d}x=\frac{3}{2}\log\left(x^2-x+1\right)+\frac{1}{\sqrt{3}}\tan^{-1}\frac{2x-1}{\sqrt{3}}+C$ then $f(x)$ is equal to

(A) $3x$

(B) $3x-4$

(C) $3x-1$

(D) $\frac{1}{1+x^2}$

Q.4

If $f(x)=\sqrt{x^2+4}$ then $\int\frac{f(x)}{x^6}\,\mathrm{d}x$ is equal to

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(A) $= \frac{1}{240x^5}\left[5x^2(f(x))^3 - 3(f(x))^5\right] + C$

(B) $\frac{(f(x))^3\left((f(x))^2-4\right)}{6x^5} + C$

(C) $\frac{(f(x))^2\left((f(x))^3-2\right)}{20x^4} + C$

(D) $\frac{(f(x))^3\left(x^2-6\right)}{6x^5} + C$

Q.5

If $I_n = \int \cot^n x \, dx$, and $I_0 + I_1 + 2\left(I_2 + \ldots + I_8\right) + I_9 + I_{10} = A\left(u + \frac{u^2}{2} + \ldots + \frac{u^9}{9}\right)$, where $u = \cot x$ then

(A) $A = 1$

(B) $A = -1$

(C) $A = 1/2$

(D) $A = -1/2$

Q.6

$f(x) = \int \frac{dx}{\sin^4 x}$ is a

(A) polynomial of degree 3 in $\cot x$

(B) polynomial of degree 4 in $\cot x$

(C) polynomial of degree 4 in $\sin x$

(D) polynomial of degree 3 in $\tan x$

Q.7

If $\int f(x)dx = \phi(x)$ i.e. ϕ is a function such that $\phi'(x) = f(x)$, then $\int x^9 f\left(x^5\right) dx$ is equal to

(A) $\frac{1}{5}\left[x^5\phi\left(x^5\right) - \int x^4\phi\left(x^5\right) dx\right] + C$

(B) $\frac{1}{5}x^5\phi\left(x^5\right) - 5\int x^5\phi\left(x^5\right) dx + C$

(C) $\frac{1}{5}x^5\phi\left(x^5\right) - \int x^4\phi\left(x^5\right) dx + C$

(D) $\frac{1}{5}\left[x^5\phi\left(x^5\right) - \int x^5\phi\left(x^5\right) dx\right] + C$

Q.8

$\int \frac{dx}{\cos x - \sin x}$ is equal to

(A) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2} - \frac{3\pi}{8}\right)\right| + C$

(B) $\frac{1}{\sqrt{2}}\log\left|\cot\frac{x}{2}\right| + C$

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(C) $\frac{1}{\sqrt{2}} \log \left|\tan \left(\frac{x}{2}-\frac{\pi}{8}\right)\right|+C$

(D) $\frac{1}{\sqrt{2}} \log \left|\tan \left(\frac{x}{2}+\frac{3 \pi}{8}\right)\right|+C$

Q.9

The value of $\sqrt{2} \int \frac{\sin x}{\sin \left(x-\frac{\pi}{4}\right)}$ is

(A) $x+\log \left|\cos \left(x-\frac{\pi}{4}\right)\right|+C$

(B) $x-\log \left|\sin \left(x-\frac{\pi}{4}\right)\right|+C$

(C) $x+\log \left|\sin \left(x-\frac{\pi}{4}\right)\right|+C$

(D) $x-\log \left|\cos \left(x-\frac{\pi}{4}\right)\right|+C$

Q.10

If $\int \frac{dx}{x+x^7}=p(x)$ then, $\int \frac{x^6}{x+x^7} dx$ is equal to

(A) $\log |x|-p(x)+C$

(B) $\log |x|+p(x)+C$

(C) $x-p(x)+C$

(D) $x+p(x)+C$

Q.11

$\int \frac{\sin^8 x-\cos^8 x}{1-2 \sin^2 x \cos^2 x} dx=$

(A) $\sin 2x+C$

(B) $-\frac{1}{2} \sin 2x+C$

(C) $\frac{1}{2} \sin 2x+C$

(D) $-\sin 2x+C$

Q.12

$\int (1-\cos x) \operatorname{cosec}^2 x dx=$

(A) $\tan \frac{x}{2}+C$

(B) $\cot \left(\frac{x}{2}\right)+C$

(C) $\frac{1}{2} \tan \frac{x}{2}+C$

(D) $2 \tan \frac{x}{2}+C$

Q.13

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Let $f(x) = \int e^x(x-1)(x-2)dx$
Then f decreases in the interval

(A) $(-\infty, -2)$

(B) $(-2, -1)$

(C) $(1, 2)$

(D) $(2, +\infty)$

Q.14

If $l^r(x)$ means $\log \log \log \ldots .x$, the log being repeated r times. then $\int \left\{\mathrm{x}l(\mathrm{x})l^2(\mathrm{x})l^3(\mathrm{x})\ldots l^{\mathrm{r}}(\mathrm{x})\right\}^{-1}\ \mathrm{dx}$ is equal to

(A) $l^{\mathrm{r}+1}(\mathrm{x}) + \mathrm{C}$

(B) $\frac{l^{\mathrm{r}+1}(\mathrm{x})}{\mathrm{r}+1} + \mathrm{C}$

(C) $l^{\mathrm{r}}(\mathrm{x}) + \mathrm{C}$

(D) None

Q.15

If $\int \frac{dx}{f(x)} = \log\{f(x)\}^2 + c$, then what is $f(x)$ equal to ?

(A) $2\mathrm{x} + \alpha$

(B) $x + \alpha$

(C) $\frac{x}{2} + \alpha$

(D) $x^2 + \alpha$

Answers & Solutions

Q.1 Answer:

$2x\sqrt{1+e^x} - 4\sqrt{1+e^x} - 2\log\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right| + C$

Solution:

First show that $f(x) = e^x$

$$\text{Let } I = \int \frac{xe^x}{\sqrt{1+e^x}}dx$$
$$= 2x\sqrt{1+e^x} - 2I_1$$

Put $1 + e^x = t^2$,

so that $I_1 = \int \frac{t2t}{t^2-1}dt = 2\int\left[1 + \frac{1}{t^2-1}\right]dt$

$= 2\left[t + \frac{1}{2}\log\left|\frac{t-1}{t+1}\right|\right]$

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$= 2\sqrt{1+e^x} + \log\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right|$

Thus, $I = 2x\sqrt{1+e^x} - 4\sqrt{1+e^x} - 2\log\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right| + C$

Q.2 Answer:

$C - \frac{1}{(1+x^2)}\tan^{-1}x + \frac{1}{2}\tan^{-1}x + \frac{x}{2(1+x^2)}$

Solution:

First show that $f(x) = 2x$.

Now, $I = \int \frac{f(x)\tan^{-1}x}{(1+x^2)^2}dx = \int \frac{2x}{(1+x^2)^2}\tan^{-1}x dx$

Put $x = \tan\theta$**, so that**

$I = \int \frac{2\tan\theta}{\sec^4\theta}\theta\sec^2\theta d\theta$

$= \int \theta\sin(2\theta)d\theta$

$= -\frac{1}{2}\theta\cos 2\theta + \frac{1}{4}\sin 2\theta + C$

$= C - \frac{1}{2}\theta\frac{1-\tan^2\theta}{1+\tan^2\theta} + \frac{1}{2}\frac{\tan\theta}{1+\tan^2\theta}$

$= C - \frac{1}{2}\frac{1-x^2}{1+x^2}\tan^{-1}x + \frac{1}{2}\frac{x}{1+x^2}$

$= C - \frac{1}{1+x^2}\tan^{-1}x + \frac{1}{2}\tan^{-1}x + \frac{1}{2}\frac{x}{1+x^2}$.

Q.3 Answer:

$3x - 1$

Solution:

Differentiating both the sides w.r.t. x**, we get** $\frac{f(x)}{x^2-x+1} = \frac{3}{2}\frac{2x-1}{x^2-x+1} + \frac{1}{\sqrt{3}}\frac{1}{1+\frac{(2x-1)^2}{3}}\left(\frac{2}{\sqrt{3}}\right)$

$= \frac{3}{2}\frac{2x-1}{x^2-x+1} + \frac{1}{2(x^2-x+1)}$

$\Rightarrow f(x) = 3x - 1$

Q.4 Answer:

$= \frac{1}{240x^5}\left[5x^2(f(x))^3 - 3(f(x))^5\right] + C$

Solution:

$I = \int \frac{\sqrt{x^2+4}}{x^6}dx = \int \frac{\sqrt{1+4/x^2}}{x^2}\frac{dx}{x^3}$

Put $\frac{1}{x^2} = t$**, so that**

$I = \int t\sqrt{1+4t}\frac{(-1)}{2}dt$

$= -\frac{1}{8}\int(4t+1-1)\sqrt{1+4t}dt$

$= -\frac{1}{8}\left[(4t+1)^{5/2}\left(\frac{2}{5}\right)\left(\frac{1}{4}\right) - \left(\frac{2}{3}\right)\left(\frac{1}{4}\right)(1+4t)^{3/2}\right] + C$

$= \frac{1}{48}\left(1+\frac{4}{x^2}\right)^{3/2} - \frac{1}{80}\left(1+\frac{4}{x^2}\right)^{5/2} + C$

$= \frac{1}{240x^5}\left[5x^2(f(x))^3 - 3(f(x))^5\right] + C$

Q.5 Answer:

$A = -1$

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Solution:

$I_n + I_{n+2} = \int \cot^n x \operatorname{cosec}^2 x dx = -\frac{1}{n+1} \cot^{n+1} x$

Now,

$I_0 + I_1 + 2(I_2 + I_3 + \cdots + I_8) + I_9 + I_{10}$

$= (I_0 + I_2) + (I_1 + I_3) + (I_2 + I_4)$

$+ \cdots + (I_7 + I_9) + (I_8 + I_{10})$

$= -\left(u + \frac{1}{2}u^2 + \frac{1}{3}u^3 + \cdots + \frac{1}{9}u^9\right)$

where $u = \cot x$.

Thus, $A = -1$

Q.6 Answer:

polynomial of degree 3 in $\cot x$

Solution:

$f(x) = \int \left(1 + \cot^2 x\right) \operatorname{cosec}^2 x dx$

$= -\left(\cot x + \frac{1}{3}\cot^3 x\right) + C$

which is a polynomial of degree 3.

Q.7 Answer:

$\frac{1}{5}x^5 \phi\left(x^5\right) - \int x^4 \phi\left(x^5\right) dx + C$

Solution:

Let $x^5 = t$**, so** $\int x^9 f\left(x^5\right) dx$

$= \frac{1}{5}\int t f(t) dt$

$= \frac{1}{5}\left[t \int f(t) dt - \int \phi(t) dt\right]$

$= \frac{1}{5}\left[x^5 \phi\left(x^5\right) - 5\int \phi\left(x^5\right) x^4 dx\right] + C$

$= \frac{1}{5}x^5 \phi\left(x^5\right) - \int x^4 \phi\left(x^5\right) dx + C$

Q.8 Answer:

$\frac{1}{\sqrt{2}} \log\left|\tan\left(\frac{x}{2} + \frac{3\pi}{8}\right)\right| + C$

Solution:

$\int \frac{dx}{\cos x - \sin x} = \int \frac{dx}{\cos x + \cos\left(\frac{\pi}{2} + x\right)}$

$= \frac{1}{2}\int \frac{dx}{\cos(\pi/4)\cos(x + \pi/4)}$

$= \frac{1}{\sqrt{2}}\int \sec\left(x + \frac{\pi}{4}\right) dx$

$= \frac{1}{\sqrt{2}} \log\left|\tan\left(\frac{x}{2} + \frac{\pi}{8} + \frac{\pi}{4}\right)\right| + C$

$= \frac{1}{\sqrt{2}} \log\left|\tan\left(\frac{x}{2} + \frac{3\pi}{8}\right)\right| + C$

Q.9 Answer:

$x + \log\left|\sin\left(x - \frac{\pi}{4}\right)\right| + C$

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Solution:

Take $\alpha = \frac{\pi}{4}$, so $\int \frac{\sin x}{\sin\left(x-\frac{\pi}{4}\right)} dx$
$= \cos\frac{\pi}{4}x + \sin\frac{\pi}{4}\log\left|\sin\left(x-\frac{\pi}{4}\right)\right| + \text{Const}$
$= \frac{1}{\sqrt{2}}\left[x + \log\left|\sin\left(x-\frac{\pi}{4}\right)\right|\right] + \text{Const}$

Q.10 Answer:

$\log|x| - p(x) + C$

Solution:

$\int \frac{x^6}{x+x^7} dx = \int \frac{1+x^6-1}{x(1+x^6)} dx = \int \frac{1}{x} dx - \int \frac{dx}{x+x^7}$
$= \log|x| - p(x) + C$

Q.11 Answer:

$-\frac{1}{2}\sin 2x + C$

Solution:

Now, $\sin^8 x - \cos^8 x$
$= \left(\sin^4 x - \cos^4 x\right)\left(\sin^4 x + \cos^4 x\right)$
$= \left(\sin^2 x - \cos^2 x\right)$
$\left(\sin^4 x + \cos^4 x + 2\sin^2 x\cos^2 x - 2\sin^2 x\cos^2 x\right)$
$= \left(\sin^2 x - \cos^2 x\right)\left(\left(\sin^2 x + \cos^2 x\right)^2 - 2\sin^2 x\cos^2 x\right)$
$= \left(\sin^2 x - \cos^2 x\right)\left(1 - 2\sin^2 x\cos^2 x\right)$
$\therefore \mathrm{I} = -\int\left(\cos^2 \mathrm{x} - \sin^2 \mathrm{x}\right)\mathrm{dx}$
$= -\int \cos 2x \mathrm{dx} = -\frac{1}{2}\sin 2\mathrm{x} + \mathrm{C}$

Q.12 Answer:

$\tan\frac{x}{2} + C$

Solution:

$\mathrm{I} = \int(1-\cos \mathrm{x})\,\mathrm{cosec}^2\,\mathrm{xdx}$
$= \int \mathrm{cosec}^2\, x dx - \int \mathrm{cosec}\, x \cot x dx$
$= -\cot x + \mathrm{cosec}\, x + C$
$= \frac{1-\cos x}{\sin x} + C = \frac{2\sin^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}} + C = \tan\frac{x}{2} + C$

Q.13 Answer:

$(1, 2)$

Solution:

$f(x) = \int e^x(x-1)(x-2)dx$
For decreasing function, $\mathrm{f}'(\mathrm{x}) < 0 \Rightarrow \mathrm{e}^{\mathrm{x}}(\mathrm{x}-1)(\mathrm{x}-2) < 0$
$\Rightarrow (\mathrm{x}-1)(\mathrm{x}-2) < 0$

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$\Rightarrow 1 < x < 2$

$e^x > 0 \forall x \in R$

Q.14 Answer:

$l^{r+1}(x) + C$

Solution:

Putting $l^{r+1}(x) = t$ **and** $\frac{1}{xl^2(x)l^3(x)....l^r(x)}dx = dt$

we get, $\int \frac{1}{xl^2(x)l^3(x)....l^r(x)}$

$= \int 1.dt = t + C = l^{r+1}(x) + C$

Q.15 Answer:

$\frac{x}{2} + \alpha$

Solution:

We check from the given options one by one. Options (1), (2) and (4) do not satisfy. We check option (3).

Let $f(x) = \frac{x}{2} + \alpha$

$\therefore \int \frac{dx}{\frac{x}{2}+\alpha} = \int \frac{2dx}{(x+2\alpha)}$

$= 2\log(x + 2\alpha) + c_1 = \log(x + 2\alpha)^2 + c_1$

$= \log\left(\frac{x}{2} + \alpha\right)^2 + \log 2^2 + c_1$

$= \log\left(\frac{x}{2} + \alpha\right)^2 + c$

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